PREMIUM RESOURCE · AI & FINANCE AUTOMATION

Power BI Finance Dashboard Starter Kit — Setup Guide

Complete setup guide for building a finance dashboard in Power BI Desktop — including data model design, 15 DAX measures, KPI card specifications, drill-through design, and data source connection instructions.

Setup Guide PDF | 15 DAX Measures | KPI Card Library | Data Source Guide

15 KPI measures	**4** report pages	**DAX** ready to copy	**Excel** data model

1 · What This Kit Includes

This starter kit gives you everything you need to build a professional finance dashboard in Power BI Desktop. The kit has two components: this PDF guide, and the Excel data model file (**power-bi-finance-data-model.xlsx**).

Component	What it contains	File
This PDF guide	Step-by-step setup, data model diagram, all 15 DAX measures, KPI card design specs, drill-through instructions, and data source connection guide	power-bi-finance-dashboard-guide.pdf
Excel data model	Pre-structured sample data: fact table (transactions), date table, accounts table, cost centres table, and budget table — ready to connect to Power BI	power-bi-finance-data-model.xlsx

BEFORE YOU START

You need **Power BI Desktop** (free download from microsoft.com/en-us/power-platform/products/power-bi/desktop). This guide is written for Power BI Desktop on Windows. All DAX measures and connection steps work with the free version — no Power BI Pro licence is required to build the dashboard locally.

2 · Data Model Design

The dashboard uses a **star schema** — one central fact table surrounded by dimension tables. This is the recommended structure for Power BI finance models because it produces the fastest query performance and the cleanest DAX measures.

Table Structure

Table	Type	Key columns	Purpose
fact_transactions	Fact	TransactionID, Date, AccountCode, CostCentreCode, Amount, TransactionType	All financial transactions — actuals
fact_budget	Fact	BudgetID, Month, AccountCode, CostCentreCode, BudgetAmount	Monthly budget by account and cost centre
dim_date	Dimension	Date, Year, Month, MonthName, Quarter, YearMonth, IsCurrentMonth, FiscalYear	Time intelligence — required for YTD and MoM
dim_accounts	Dimension	AccountCode, AccountName, Category, SubCategory, PLSection, IsRevenue, IsCost	Chart of accounts hierarchy
dim_costcentres	Dimension	CostCentreCode, CostCentreName, Department, Division, Region	Organisational hierarchy

Relationships

STAR SCHEMA RELATIONSHIPS

fact_transactions[Date] → **dim_date[Date]** — Many-to-one, single direction, active

fact_transactions[AccountCode] → **dim_accounts[AccountCode]** — Many-to-one, single direction, active

fact_transactions[CostCentreCode] → **dim_costcentres[CostCentreCode]** — Many-to-one, single direction, active

fact_budget[AccountCode] → **dim_accounts[AccountCode]** — Many-to-one, single direction, active

fact_budget[Month] → **dim_date[YearMonth]** — Many-to-one, single direction, active

CRITICAL: MARK THE DATE TABLE

After loading dim_date, right-click the table in the Fields pane → Mark as date table → select the Date column. Without this step, time intelligence functions (YTD, MoM, etc.) will not work correctly.

3 · Connecting Your Data Source

The Excel data model file provides sample data to get started. For production use, replace it with your own data source using one of the connection methods below.

Option A — Excel / CSV (Quickest Start)

1. **Open Power BI Desktop → Home → Get Data → Excel Workbook**
   Navigate to power-bi-finance-data-model.xlsx and click Open.
2. **Select all five tables in the Navigator window**
   Check fact_transactions, fact_budget, dim_date, dim_accounts, dim_costcentres. Click Transform Data.
3. **In Power Query Editor — verify column data types**
   Date columns must be Date type. Amount must be Decimal Number. Code columns must be Text. Fix any mismatches before loading.
4. **Click Close & Apply**
   Power BI loads all tables and creates the data model. Then set up relationships as described in Section 2.

Option B — SQL Server / Azure SQL

1. **Home → Get Data → SQL Server**
   Enter your server name and database name. Use DirectQuery mode for large datasets (>5M rows), Import mode for smaller datasets where speed is the priority.
2. **Write custom SQL queries for each table**
   Replace the Excel tables with SQL views or queries that return the same column structure. The DAX measures require no changes if the column names match.
3. **Set up scheduled refresh (Power BI Service)**
   Publish the report to Power BI Service → Dataset settings → Scheduled refresh. Requires a gateway for on-premises SQL Server.

Option C — SAP / Oracle / Dynamics (via ODBC)

1. **Home → Get Data → ODBC**
   Select your pre-configured ODBC data source name (DSN). If no DSN exists, set it up in Windows ODBC Data Source Administrator first.
2. **Build extraction queries matching the data model structure**
   For SAP: extract from GL account line items (BSEG/BKPF). For Oracle Financials: extract from GL_JE_LINES. Map to the fact_transactions structure.
3. **Use Power Query to reshape and clean**
   SAP and Oracle data often needs reshaping — pivot/unpivot, column renaming, and data type correction. Do all transformation in Power Query, not in DAX.

> **BEST PRACTICE: NEVER TRANSFORM IN DAX**
> All data cleaning, column renaming, filtering, and reshaping should happen in Power Query (before the data loads). DAX should only calculate — not clean. This keeps your model fast and your measures simple.

4 · DAX Measures — All 15 KPI Calculations

Copy each measure into Power BI Desktop: select the target table in the Fields pane → Table tools → New measure → paste the DAX. All measures reference the table and column names from the Excel data model. If you rename any columns, update the references here.

> **MEASURE ORGANISATION**
>
> Create a dedicated _**Measures** table (Home → Enter Data → name it _Measures → Load) and store all measures there. This keeps them separate from data tables and makes the field pane easier to navigate. The underscore prefix ensures it sorts to the top.

Group 1 — Revenue Measures

KPI 01 — ACTUAL REVENUE

```
Actual Revenue =
CALCULATE(
    SUM(fact_transactions[Amount]),
    dim_accounts[IsRevenue] = TRUE,
    fact_transactions[TransactionType] = "Actual"
)
```

KPI 02 — REVENUE YTD

```
Revenue YTD =
CALCULATE(
    [Actual Revenue],
    DATESYTD(dim_date[Date])
)
```

KPI 03 — REVENUE VS BUDGET

```
Revenue vs Budget =
VAR BudgetRev =
    CALCULATE(
        SUM(fact_budget[BudgetAmount]),
        dim_accounts[IsRevenue] = TRUE
    )
RETURN
    [Actual Revenue] - BudgetRev
```

KPI 04 — REVENUE VS BUDGET %

```
Revenue vs Budget % =
VAR BudgetRev =
    CALCULATE(
        SUM(fact_budget[BudgetAmount]),
        dim_accounts[IsRevenue] = TRUE
    )
RETURN
    DIVIDE([Actual Revenue] - BudgetRev, ABS(BudgetRev))
```

KPI 05 — REVENUE MONTH-ON-MONTH CHANGE

```
Revenue MoM Change =
VAR PrevMonth =
    CALCULATE(
        [Actual Revenue],
        PREVIOUSMONTH(dim_date[Date])
    )
RETURN
    DIVIDE([Actual Revenue] - PrevMonth, ABS(PrevMonth))
```

Group 2 — Cost Measures

KPI 06 — ACTUAL COSTS

```
Actual Costs =
CALCULATE(
    SUM(fact_transactions[Amount]),
    dim_accounts[IsCost] = TRUE,
    fact_transactions[TransactionType] = "Actual"
)
```

KPI 07 — COSTS YTD

```
Costs YTD =
CALCULATE(
    [Actual Costs],
    DATESYTD(dim_date[Date])
)
```

KPI 08 — COSTS VS BUDGET

```
Costs vs Budget =
VAR BudgetCost =
    CALCULATE(
        SUM(fact_budget[BudgetAmount]),
        dim_accounts[IsCost] = TRUE
    )
RETURN
    [Actual Costs] - BudgetCost
```

Group 3 — Profit & Margin Measures

KPI 09 — GROSS PROFIT

```
Gross Profit =
[Actual Revenue] - [Actual Costs]
```

KPI 10 — GROSS MARGIN %

```
Gross Margin % =
DIVIDE([Gross Profit], [Actual Revenue])
```

KPI 11 — GROSS PROFIT YTD

```
Gross Profit YTD =
CALCULATE(
    [Gross Profit],
    DATESYTD(dim_date[Date])
)
```

Group 4 — Budget & Variance Measures

KPI 12 — TOTAL BUDGET

```
Total Budget =
SUM(fact_budget[BudgetAmount])
```

KPI 13 — OVERALL VARIANCE VS BUDGET

```
Variance vs Budget =
VAR ActualGP = [Gross Profit]
VAR BudgetGP =
    CALCULATE(
        SUM(fact_budget[BudgetAmount]),
        dim_accounts[IsRevenue] = TRUE
    )
    -
    CALCULATE(
        SUM(fact_budget[BudgetAmount]),
        dim_accounts[IsCost] = TRUE
    )
RETURN
    ActualGP - BudgetGP
```

KPI 14 — PRIOR YEAR COMPARISON

```
Revenue Prior Year =
CALCULATE(
    [Actual Revenue],
    SAMEPERIODLASTYEAR(dim_date[Date])
)

Revenue YoY Change % =
DIVIDE(
    [Actual Revenue] - [Revenue Prior Year],
    ABS([Revenue Prior Year])
)
```

KPI 15 — ROLLING 3-MONTH AVERAGE REVENUE

```
Revenue Rolling 3M Avg =
CALCULATE(
    [Actual Revenue],
    DATESINPERIOD(
        dim_date[Date],
        LASTDATE(dim_date[Date]),
        -3,
        MONTH
    )
) / 3
```

5 · KPI Card Design Specifications

Each KPI card in Power BI uses a **Card visual** or **New Card visual** (Power BI 2023+). The new card visual supports multiple fields, conditional formatting, and reference labels — use it where available. The specifications below define exactly how each card should be configured.

Card Design System

Element	Setting	Value / Guidance
Background	Card fill	White (#FFFFFF) for standard cards; Dark navy (#0D1B2A) for highlight/summary cards
Border	Card border	1pt, colour: #E2E8F0 (light grey). Left accent border: 3pt, colour: #C07A2D (amber) for primary KPIs
Value font	Callout value	Arial, Bold, 22pt, #0D1B2A for dark-on-light; #FFFFFF for light-on-dark cards
Label font	Category label	Arial, Regular, 9pt, #64748B
Trend indicator	Reference label	Use MoM or YTD measure as reference. Green (#1E6B4A) for favourable, Red (#991B1B) for adverse
Conditional formatting	Background colour	Apply to variance cards: green background for positive variance, red for negative

15 KPI Card Specifications

01
Actual Revenue
Measure: [Actual Revenue]
Format: Currency, 0 decimal places. Reference label: [Revenue MoM Change] as %

02
Revenue YTD
Measure: [Revenue YTD]
Format: Currency, 0 decimal places. Reference label: "YTD" static text

03
Revenue vs Budget (£)
Measure: [Revenue vs Budget]
Format: Currency with sign. Conditional format: green if >0, red if <0

04
Revenue vs Budget (%)
Measure: [Revenue vs Budget %]
Format: Percentage, 1 decimal. Conditional format: green if >0, red if <0

05
Revenue MoM Change
Measure: [Revenue MoM Change]
Format: Percentage, 1 decimal. Add arrow icon: ▲ green if positive, ▼ red if negative

06
Actual Costs
Measure: [Actual Costs]
Format: Currency, 0 decimal places. Reference label: [Costs vs Budget]

07
Costs YTD
Measure: [Costs YTD]
Format: Currency, 0 decimal places. Reference label: "YTD" static text

08
Costs vs Budget
Measure: [Costs vs Budget]
Format: Currency with sign. Note: for costs, negative = favourable (underspend)

09
Gross Profit

Measure: [Gross Profit]

Format: Currency, 0 decimal places. Reference label: [Gross Margin %]

10
Gross Margin %

Measure: [Gross Margin %]

Format: Percentage, 1 decimal. Add gauge visual below showing margin vs budget margin target

11
Gross Profit YTD

Measure: [Gross Profit YTD]

Format: Currency, 0 decimal places. Reference label: [Variance vs Budget]

12
Total Budget

Measure: [Total Budget]

Format: Currency, 0 decimal places. Static card — no conditional formatting needed

13
Variance vs Budget

Measure: [Variance vs Budget]

Format: Currency with sign. Highlight card — use navy background, white text. Conditional format fill

14
Revenue Year-on-Year

Measure: [Revenue YoY Change %]

Format: Percentage, 1 decimal. Reference label: [Revenue Prior Year] as context

15
Rolling 3-Month Avg Revenue

Measure: [Revenue Rolling 3M Avg]

Format: Currency, 0 decimal places. Use in line chart as a trend overlay series

6 · Report Page Layout

The dashboard is built across four report pages. Each page has a specific audience and purpose. The canvas size for all pages: **1280 × 720px** (16:9 widescreen — standard for finance packs).

Page 1 — Executive Summary

AUDIENCE: CFO / FINANCE DIRECTOR

Row 1 (top strip): Month/period slicer, YTD toggle, last refresh timestamp

Row 2 (KPI cards): KPI 01 Actual Revenue · KPI 09 Gross Profit · KPI 10 Gross Margin % · KPI 13 Variance vs Budget

Row 3 (charts): Clustered bar chart — Revenue vs Budget by month (12 months) · Line chart — Gross Profit trend with Rolling 3M Avg overlay

Row 4 (table): P&L summary by category — Revenue / Cost of Sales / Gross Profit / Opex / EBITDA — Actual vs Budget vs Prior Year

Page 2 — Revenue Analysis

AUDIENCE: FINANCE BUSINESS PARTNER / COMMERCIAL TEAM

Top KPIs: KPI 01 · KPI 02 · KPI 05 · KPI 14

Main visual: Revenue waterfall chart — Prior Year → MoM movements → Current Month

Secondary: Revenue by cost centre (stacked bar) · Revenue by account subcategory (treemap)

Slicer panel: Department / Division / Region / Account Category

Page 3 — Cost & Variance Analysis

AUDIENCE: FINANCE BUSINESS PARTNER / BUDGET HOLDERS

Top KPIs: KPI 06 · KPI 07 · KPI 08 · KPI 12

Main visual: Variance waterfall — Budget → Actual by cost category

Secondary: Costs by department (horizontal bar, sorted descending) · Budget vs Actual scatter plot by cost centre

Table: Cost detail by account with variance column — sortable, conditional colour on variance

Page 4 — Drill-Through Detail

NAVIGATION: RIGHT-CLICK ANY VISUAL → DRILL THROUGH → TRANSACTION DETAIL

This page is hidden from the navigation pane — it is accessed by drill-through only. It shows transaction-level detail for whatever the user right-clicked on.

Setup: Add a drill-through filter field (e.g. AccountCode or CostCentreCode) to the Drill through well in the Visualizations pane. Any visual on the other pages that contains this field will allow drill-through to this page.

Content: Transaction table — Date · Account · Description · Cost Centre · Amount · Type · Running total · Export to Excel button

Drill-Through Setup Steps

1. **Create Page 4 and name it "Transaction Detail"**
   In the Pages pane, right-click the page → Hide page. It will still be accessible via drill-through but won't appear in the navigation.

2. **Add a drill-through filter**
   On Page 4, open Visualizations → Drill through well. Drag CostCentreCode from the Fields pane into the Drill through well.

3. **Add the transaction table to Page 4**
   Add a Table visual with: dim_date[Date], dim_accounts[AccountName], fact_transactions[Amount], dim_costcentres[CostCentreName], fact_transactions[TransactionType]. Sort by Date descending.

4. **Test the drill-through**
   Go to Page 2 or 3. Right-click any bar or data point. Select Drill through → Transaction Detail. The page should appear filtered to the selected cost centre or account.

7 · Replacing Sample Data with Your Own

The Excel data model contains 24 months of sample data across 6 cost centres and 30 account codes. When you are ready to use your own data, follow these steps to replace the sample data without rebuilding the model.

1. **Export your chart of accounts to match the dim_accounts structure**
   Required columns: AccountCode (text, unique), AccountName, Category, SubCategory, PLSection, IsRevenue (TRUE/FALSE), IsCost (TRUE/FALSE). Every account in your transaction data must exist in this table.

2. **Export your transaction data to match the fact_transactions structure**
   Required columns: TransactionID (unique), Date (date), AccountCode (must match dim_accounts), CostCentreCode (must match dim_costcentres), Amount (positive for revenue, positive for cost — the IsRevenue/IsCost flags control the sign in DAX), TransactionType ("Actual").

3. **Export your budget data to match the fact_budget structure**
   Required columns: BudgetID (unique), Month (YYYY-MM format, e.g. "2026-01"), AccountCode, CostCentreCode, BudgetAmount. One row per account per cost centre per month.

4. **Paste your data into the corresponding sheets in the Excel file**
   Overwrite the sample data rows — keep the header row unchanged. Column names must match exactly. Save the Excel file.

5. **In Power BI Desktop: Home → Refresh**
   Power BI reloads all tables from the Excel file. All visuals and DAX measures update automatically. No rebuilding required.

DATE TABLE NOTE

The dim_date table in the Excel file covers 2023–2027. If your data extends beyond this range, extend the date table by adding rows at the top or bottom. The columns and data types must remain unchanged. Alternatively, replace dim_date with a calculated DAX date table: `CALENDAR(DATE(2020,1,1), DATE(2030,12,31))` with calculated columns for Year, Month, MonthName, etc.

8 · Publishing & Sharing

Option	How	Requires
Share as PDF	File → Export → Export to PDF. Each report page becomes one PDF page.	Nothing — works with free Power BI Desktop
Publish to Power BI Service	Home → Publish → select your workspace. Share the report link with colleagues.	Power BI Pro licence (£9.40/user/month) or Premium Per User
Embed in Teams	In Teams, add a Power BI tab to a channel → select your published report.	Power BI Pro + Microsoft Teams
Scheduled refresh	In Power BI Service → Dataset → Scheduled refresh. Set daily or weekly.	Power BI Pro + on-premises gateway (for Excel/SQL on local network)
Export to Excel	In any table visual → right-click → Export data. Exports the filtered view.	Nothing — available in both Desktop and Service